

Agent vs OTel: ingest to Lucene shards

Two philosophies land in the same Elasticsearch storage layer on Serverless project `my-observability-project-dce7f4` . Writes go to a data stream name. Elasticsearch creates hidden backing indices. Lucene segments for those indices are what actually sit on disk.

2	9	3	ECS vs OTel
Ingest philosophies	Workshop data streams	Rolled to 000002	Document contract

Path through Elastic

Left to right is time. The fork is ownership and schema, not a different cluster. Both paths resolve an index template, attach to a data stream, and flush Lucene segments into the current write backing index.

	1. Origin	2. Shipper	3. Elastic intake	4. Ingest pipeline	5. Template match	6. Data stream	7. Backing index	8. On disk
Agent	Host files <code>/var/log/secure</code> and <code>/var/log/messages</code> on aks-sre-01..03	Elastic Agent Fleet policy sre-01-workshop · System integration · Beat receivers · ECS	Fleet Server Enrolled agents bulk-index to Elasticsearch over the Fleet connection	ECS pipelines run logs-system.auth-2.22.4 · logs-system.syslog-2.22.4 · metrics-system.*-2.22.4	Integration templates logs-system.* / metrics-system.* · type-dataset-namespace naming	Write alias logs-system.auth-default · logs-system.syslog-default · metrics-system.*-default	Generation 1 .ds-logs-system.auth-default-2026.08.18-000001 (write = 000001)	logsdb Lucene shards Serverless allocates shards; segments land in the write backing index
OTel	Apps + factory OTLP logs, metrics, traces from well-data-api and siblings	EDOT collector elastic-agent otel mode · OTLP :4317/:4318 · batch + resource processors	Managed OTLP ingest.us-east-1.aws.elastic.cloud — not APM Server OTLP	ECS pipelines skipped logs@default-pipeline only · traces and metrics have no default_pipeline	OTel templates logs-*.otel-* · metrics-otel@template · traces-*.otel-*	Write alias logs-workshop.otel.otel-default · traces-workshop.otel.otel-default · metrics-workshop.otel.otel-default	Rolled to gen 2 000001 holds history · 000002 is the write index after POST _rollover	logsdb or TSDS shards Logs/traces: logsdb · Metrics: time_series (~2h look-back)

Where the shard actually is
Serverless does not expose `_cat/shards` or `index.number_of_shards` . Each backing index still has Lucene shards. The write index receives new documents; a rollover freezes the previous generation and opens the next. Search the stream name, never `.ds→*` directly.

Features and operating model

Same Elastic Agent binary family, two operating models. This workshop runs Fleet-managed Agent beside Agent in `otel` mode — the EDOT collector. Choose on ownership and schema, not vendor preference. Both can coexist.

Operating model		
	Fleet-managed Elastic Agent	EDOT (Agent in otel mode)
Who owns rollout	Central SRE / platform team	App or platform team plus their CM tool (git, Helm)
Config artifact	Fleet policy (UI / API). Policy is the artifact; binary is cattle	OTel YAML: receivers / processors / exporters
Central policy push	Yes — Fleet enrolls and updates agents	No — otel mode cannot enroll in Fleet
Central monitoring in Fleet	Yes	Planned
Intake on this Serverless project	Fleet Server → Elasticsearch bulk	Managed OTLP Endpoint (not APM Server OTLP)
Support boundary	Fleet integrations and ECS pipelines are supported	EDOT SDKs only with Managed OTLP or Agent Gateway — never APM Server OTLP
Change later	Allowed. Dashboards, pipelines, and runbooks all change	Allowed. Same cost: new streams, new mappings, new on-call

Features		
	Fleet-managed Elastic Agent	EDOT (Agent in otel mode)
Beat receivers (ECS)	Yes — System integration in this workshop	Available on the binary; this otel path does not use them
OTLP receivers	No on policy sre-01-workshop	Yes — gRPC 4317 / HTTP 4318
Host logs and metrics	Yes — logs-system.* / metrics-system.*	If configured (filelog / hostmetrics); this lab uses app OTLP + syslog via OTel
Distributed traces	No on the System integration	Yes — traces-*.otel-*
Curated integrations and Kibana assets	Yes — Fleet packages, dashboards, @custom pipelines	OTel input packages (preview) / YAML; no auto asset install

	Fleet-managed Elastic Agent	EDOT (Agent in otel mode)
Ingest pipelines	Yes — ECS pipelines on the backing index	OTel-native docs skip ECS pipelines (logs@default-pipeline only)
Document schema	ECS (event.action, host.name, system.cpu.*)	OTel semantic conventions (resource.attributes, severity_text)
Elastic Defend / Cloud Security	Yes	No
Streams in this workshop	logs-system.auth syslog-default, metrics-system.*-default	logs-*.otel-*, metrics-workshop.otel.otel-default, traces-workshop.otel.otel-default

Rule of thumb
Central SRE owns host, security, and packaged integrations → Fleet-managed Agent. App team already lives in OTel Collector YAML / Helm → EDOT in `otel` mode (Managed OTLP on this project). Do not point EDOT SDKs at APM Server OTLP.

This project, after rollover

Data streams → backing indices				counts at 18 Aug 2026 16:39 UTC	
Path	Data stream	Ingest pipeline	Index mode	Write gen	Docs
● Agent	logs-system.auth-default	logs-system.auth-2.22.4	logsdb	000001	4,970
● Agent	logs-system.syslog-default	logs-system.syslog-2.22.4	logsdb	000001	4,970
● Agent	metrics-system.cpu-default	metrics-system.cpu-2.22.4	time_series	000001	607
● Agent	metrics-system.memory-default	metrics-system.memory-2.22.4	time_series	000001	606
● OTel	logs-workshop.otel.otel-default	logs@default-pipeline	logsdb	000002	3,067
● OTel	logs-system.auth.otel-default	logs@default-pipeline	logsdb	000001	1,036
● OTel	traces-workshop.otel.otel-default	—	logsdb	000002	3,060
● OTel	metrics-workshop.otel.otel-default	—	time_series	000001	186
● Lab 1	logs-workshop.platform-default	logs-workshop.platform	logsdb	000002	3,072

Agent: pipeline is the contract

Fleet ships ECS. The System integration sets `index.default_pipeline` on the backing index. Auth lines become `event.action`, `process.name`, `user.name` before Lucene sees the document.

Metrics follow the same idea with TSDS: `metrics-system.cpu-2.22.4` runs, then the document is a time-series Lucene shard keyed on host dimensions plus `@timestamp`.

OTel: mapping is the contract

Managed OTLP writes OTel semantic conventions into `*.otel-*` streams. There is no System grok pipeline. Logs get `logs@default-pipeline` only. Traces and OTel metrics skip ingest pipelines entirely.

That is why Agent and EDOT dashboards cannot share the same fields: different streams, different mappings, different on-call runbooks — same disk abstraction underneath.

Lab 1 bulk path `scripts/ingest.py` is a third writer: it targets `logs-workshop.platform-default` with pipeline `logs-workshop.platform` (JSON/grok → ECS). Same landing: data stream → `.ds-...-000002` → logsdb Lucene shards. Source: Serverless project my-observability-project-dce7f4 after the 18 Aug 2026 rebuild and rollover. From the SRE-01 workshop canvas.